Gumbel-max Watermark

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Lemma 1.

If $X \sim U(0,1), Y = -\ln(X)$*, then* $Y \sim \mathrm{Exp}(1)$

Proof.
This is clear by manipulating their CDF.

□

Now

$$\arg\max_w \frac{\log U_w}{P_w} \iff \arg\min_w \frac{-\log U_w}{P_w} \iff \arg\min_w \frac{Exp(1)}{P_w}$$

Note (by property of Exp or Gamma, as we learned in stat 450),

$$Y_w := \frac{Exp(1)}{P_w} \sim Exp(\lambda = P_w)$$

By stat 333,

Lemma 2.

Let $\{X_i\}_{i=1}^n$ *be a sequence of independent rvs where* $X_i \sim \mathrm{EXP}(\lambda_i),\ i = 1, 2, \dots, n$*. For* $j \in \{1, 2, \dots, n\}$*,* $\mathbb{P}(X_j = \min\{X_1, X_2, \dots, X_n\}) = \frac{\lambda_j}{\sum_{i=1}^n \lambda_i}$

Proof.

$$\begin{aligned}
&\mathbb{P}(X_j = \min\{X_1, X_2, \dots, X_n\}) \\
=&\mathbb{P}(X_j < X_1, X_j < X_2, \dots, X_j < X_n) \\
=&\int_0^\infty \mathbb{P}(X_j < X_1, X_j < X_2, \dots, X_j < X_n \mid X_j = x)\lambda_j e^{-\lambda_j x}\, dx \\
=&\int_0^\infty \mathbb{P}(X_1 > x, X_2 > x, \dots, X_n > x)\lambda_j e^{-\lambda_j x}\, dx \quad (\text{since } X_j \text{ is independent of } \{X_i\}_{i\neq j}) \\
=&\int_0^\infty \mathbb{P}(X_1 > x)\mathbb{P}(X_2 > x)\cdots\mathbb{P}(X_n > x)\lambda_j e^{-\lambda_j x}\, dx \\
=&\int_0^\infty e^{-\lambda_1 x}e^{-\lambda_2 x}\cdots e^{-\lambda_n x}\lambda_j e^{-\lambda_j x}\, dx \\
=&\int_0^\infty \lambda_j e^{-(\sum_{i=1}^n \lambda_i)x}\, dx \\
=&\frac{\lambda_j}{\sum_{i=1}^n \lambda_i}\int_0^\infty \left(\sum_{i=1}^n \lambda_i\right) e^{-(\sum_{i=1}^n \lambda_i)x}\, dx \\
=&\frac{\lambda_j}{\sum_{i=1}^n \lambda_i}
\end{aligned}$$

□

Therefore

$$\mathbb{P}(Y_w \text{ is the minimum}) = \frac{P_w}{1} = P_w$$

and $\{\arg\min_{k\in W} Y_k = w\} = \{Y_w\ is\ minimum\}$ as events, so we are done.

Proof of Theorem 2.1

Consider equation (30):

- Consider the 1^{st} inequality.
  Observe

$$\mathbb{E}_1[T_h(Y_{1:n})] = 1 \cdot \mathbb{P}_1(T_h = 1) + 0 \cdot \mathbb{P}_1(T_h = 0) = \mathbb{P}_1(T_h = 1)$$

  And so

$$1 - \mathbb{E}_1[T_h(Y_{1:n})] = \mathbb{P}_1(T_h = 0)$$

  Write

$$\mathbb{P}_1(T_h = 0) = \mathbb{P}_1\left(\sum_{t=1}^{n} h(Y_t) < \gamma_{n,\alpha}\right) = \mathbb{P}_1\left(\sum_{t=1}^{n} -h(Y_t) > -\gamma_{n,\alpha}\right)$$

  If $\sum_{i=1}^{n} -h(Y_t) > -\gamma_{n,\alpha}$, then $\theta \sum_{i=1}^{n} -h(Y_t) > -\theta\gamma_{n,\alpha}$, thus

$$\exp\left(\sum_{i=1}^{n} -\theta h(Y_t)\right) > \exp\left(-\theta\gamma_{n,\alpha}\right)$$

  By Markov, for nonnegative Z, $a > 0$, $\mathbb{P}(Z \geq a) \leq \frac{\mathbb{E}[Z]}{a}$, so

$$\mathbb{P}_1\left(\sum_{i=1}^{n} -h(Y_t) > -\gamma_{n,\alpha}\right) \leq \frac{\mathbb{E}_1[\exp\left(\sum_{i=1}^{n} -\theta h(Y_t)\right)]}{\exp\left(-\theta\gamma_{n,\alpha}\right)}$$

  (In discrete case, note $\mathbb{P}(Z > a) \leq \mathbb{P}(Z \geq a)$. In cts case, they are equal.)
- Consider the 2^{nd} inequality
  Since

$$\mathbb{E}_{H_1} \exp\left(\sum_{t=1}^{n} -\theta h(Y_t)\right) = \mathbb{E}_{H_1}\left[\prod_{t=1}^{n} \exp(-\theta h(Y_t))\right]$$

  By tower rule,

$$\begin{aligned}\mathbb{E}_{H_1}\left[\prod_{t=1}^{n} \exp(-\theta h(Y_t))\right] &= \mathbb{E}_{H_1}\left[\mathbb{E}_{H_1}\left[\prod_{t=1}^{n} \exp(-\theta h(Y_t)) \mid (w_{1:(t-1)}, \zeta_{1:(t-1)})\right]\right] \\ &= \mathbb{E}_{H_1}\left[\left(\prod_{t=1}^{n-1} \exp(-\theta h(Y_t))\right)\mathbb{E}_{H_1}[\exp(-\theta h(Y_n)) \mid (w_{1:(t-1)}, \zeta_{1:(t-1)})]\right]\end{aligned}$$

  By **working hypothesis 2.1** (here P_t is known and ζ_t independent), this is

$$\mathbb{E}_{H_1}\left[\left(\prod_{t=1}^{n-1} \exp(-\theta h(Y_t))\right)\mathbb{E}_{1,P_n}[\exp(-\theta h(Y_n))]\right]$$

  Note

$$\mathbb{E}_{1,P_n}[\exp(-\theta h(Y_n))] = \exp(\log \phi_{P_n,h}(\theta)) \leq \exp(\log \phi_{\mathcal{P},h}(\theta))$$

  where $\phi_{\mathcal{P},h}(\theta) = \sup_{P\in\mathcal{P}} \phi_{P,h}(\theta) = \sup_{P\in\mathcal{P}} \mathbb{E}_{1,P}[\exp(-\theta h(Y))]$.

And by repeatedly doing above from $t = n-1$ to $t = 1$,

$$\mathbb{E}_{H_1}\left[\prod_{t=1}^{n} \exp(-\theta h(Y_t))\right] \leq (\exp(\log \phi_{\mathcal{P},h}(\theta)))^n = \exp(n \cdot \log \phi_{\mathcal{P},h}(\theta))$$

For the last equality in the proof,

$$\inf_{\theta\geq 0} \exp\left(\theta\mathbb{E}_0 h(Y) + \log \phi_{\mathcal{P},h}(\theta)\right) = \exp\left(\inf_{\theta\geq 0}\left[\theta\mathbb{E}_0 h(Y) + \log \phi_{\mathcal{P},h}(\theta)\right]\right)$$

as $\exp$ is continuous and monotone increasing.

Intuition for equation (29) and its proof

To control type 1 error, for $\alpha \in (0, 1)$

$$\mathbb{P}_{H_0}\left(\sum_{t=1}^{n} h(Y_t) \geq \gamma_{n,\alpha}\right) = \alpha$$

which is

$$\mathbb{P}_{H_0}\left(\frac{1}{n}\sum_{t=1}^{n} h(Y_t) \geq \frac{\gamma_{n,\alpha}}{n}\right) = \alpha$$

Under H_0, Y_i are iid. Thus by SLLN,

$$\frac{1}{n}\sum_{t=1}^{n} h(Y_t) \xrightarrow{a.s.} \mathbb{E}_0 h(Y)$$

So we guess

$$\lim_{n\to\infty} \frac{\gamma_{n,\alpha}}{n} = \mathbb{E}_0 h(Y)$$

because otherwise, α will go to either 0 or 1.

Using contradiction, the paper shows

$$\limsup_{n\to\infty} \frac{\gamma_{n,\alpha}}{n} \leq \mathbb{E}_0 h(Y)$$

and

$$\liminf_{n\to\infty} \frac{\gamma_{n,\alpha}}{n} \geq \mathbb{E}_0 h(Y)$$

and so the limit exists and equals $\mathbb{E}_0 h(Y)$.

KL Divergence and Donsker-Varadhan

Definition 3.

Let P and Q be two probability measures on a measurable space $(\mathcal{X}, \mathcal{F})$. Assume P is absolutely continuous with respect to Q (denoted as $P \ll Q$), meaning that for any measurable set A, if $Q(A) = 0$, then $P(A) = 0$. By the Radon-Nikodym theorem, there exists a density function (or likelihood ratio) $\frac{dP}{dQ}$ such that $P(A) = \int_A \frac{dP}{dQ} dQ$.

The KL divergence of P from Q is defined as:

$$D_{\mathrm{KL}}(P\|Q) = \mathbb{E}_P\left[\log \frac{dP}{dQ}\right] = \int_{\mathcal{X}} \log\left(\frac{dP}{dQ}\right) dP$$

If P is not absolutely continuous with respect to Q, then $D_{\mathrm{KL}}(P\|Q) = \infty$.

Remark 4.

Note that every distribution is a probability measure (push forward measure): $F_X(x) = \mathbb{P} \circ X^{-1}((-\infty, x])$. By Carathéodory extension thm, CDF uniquely determines its probability measure.

Proposition 5.

We have

$$D_{\mathrm{KL}}(P\|Q) \geq 0$$

with equality if and only if $P = Q$ as probability measures.

Proof.
If $P \not\ll Q$, then, by definition,

$$D_{\mathrm{KL}}(P\|Q) = +\infty,$$

so the result is immediate. We therefore assume that $P \ll Q$, and let

$$p = \frac{dP}{dQ}.$$

Then

$$\int_{\mathcal{X}} p\,dQ = P(\mathcal{X}) = 1,$$

and, using the convention $0\log 0 = 0$ (where $x\log x$ is defined at $x = 0$ by continuous extension, so that $0\log 0 = 0$),

$$D_{\mathrm{KL}}(P\|Q) = \int_{\mathcal{X}} \log p\,dP = \int_{\mathcal{X}} p\log p\,dQ.$$

For $x \geq 0$, define

$$h(x) = x\log x - x + 1,$$

where $h(0) = 1$. For $x > 0$,

$$h'(x) = \log x.$$

Thus h is decreasing on $(0, 1)$ and increasing on $(1, \infty)$, so it has a unique minimum at $x = 1$. Since

$$h(1) = 0,$$

we have

$$x\log x - x + 1 \geq 0,$$

or equivalently,

$$x\log x \geq x - 1,$$

with equality if and only if $x = 1$.

Applying this inequality to p, we obtain

$$\begin{aligned}
D_{\mathrm{KL}}(P\|Q) &= \int_{\mathcal{X}} p\log p\,dQ \\
&= \int_{\mathcal{X}} \big(p\log p - p + 1\big)\,dQ + \int_{\mathcal{X}} (p-1)\,dQ \\
&= \int_{\mathcal{X}} \big(p\log p - p + 1\big)\,dQ \\
&\geq 0,
\end{aligned}$$

because

$$\int_{\mathcal{X}} (p-1)\,dQ = 1 - 1 = 0.$$

If $D_{\mathrm{KL}}(P|Q) = 0$, then

$$\int_{\mathcal{X}} \big(p\log p - p + 1\big)\,dQ = 0.$$

The integrand is nonnegative and vanishes only when $p = 1$. Therefore,

$$p = 1 \qquad Q\text{-almost surely.}$$

Consequently, for every $A \in \mathcal{F}$,

$$P(A) = \int_A p\, dQ = \int_A 1\, dQ = Q(A),$$

so $P = Q$.

Conversely, if $P = Q$, then

$$\frac{dP}{dQ} = 1 \qquad Q\text{-almost surely},$$

and hence

$$D_{\mathrm{KL}}(P\|Q) = \int_{\mathcal{X}} \log 1\, dP = 0.$$

Therefore,

$$D_{\mathrm{KL}}(P\|Q) \geq 0$$

with equality if and only if $P = Q$.

□

Theorem 6 (Jensen's inequality)**.**

Suppose that X is a random variable with expectation μ, and function g is convex and finite. Then

$$\mathbb{E}_X[g(X)] \geq g(\mathbb{E}_X[X])$$

with equality if and only if, for every line $a + bx$ that is a tangent to g at μ

$$\mathrm{P}_X[g(X) = a + bX] = 1.$$

https://www.math.mcgill.ca/dstephens/556-2014/Handouts/Math556-05-Inequalities.pdf

Corollary 7.

Suppose g is strictly convex and finite. Then $\mathbb{E}[g(X)] = g(\mathbb{E}[X])$ if and only if $\mathbb{P}[X = \mu] = 1$

Proof.
$\Longrightarrow$: Assume $\mathbb{E}[g(X)] = g(\mathbb{E}[X])$. By theorem, for any tangent line $L(x) = a + bx$ to g at the point μ, we have $\mathbb{P}[g(X) = a + bX] = 1$. Because g is convex and finite, it possesses at least one subgradient at μ (This is fact 9.12 and corollary 9.13 in my co463 notes). Therefore, there exists at least one line $L(x) = a + bx$ that is tangent to g at μ. This guarantees the existence of a and b such that $g(\mu) = a + b\mu$. By the definition of strict convexity, a strictly convex function intersects any of its tangent lines at exactly one point: the point of tangency. Therefore, for all $x \neq \mu$, we have a strict inequality: $g(x) > a + bx$. Consequently, the equation $g(x) = a + bx$ holds true if and only if $x = \mu$. Define the events $A = \{\omega \in \Omega \mid g(X(\omega)) = a + bX(\omega)\}$ and $B = \{\omega \in \Omega \mid X(\omega) = \mu\}$. Then $g(X(\omega)) = a + bX(\omega) \iff X(\omega) = \mu$ holds for every ω by above. That is $A = B$. Recall we know $\mathbb{P}(A) = 1$. So $\mathbb{P}(B) = 1$, which means $\mathbb{P}[X = \mu] = 1$.

$\Longleftarrow$: Assume $\mathbb{P}[X = \mu] = 1$. Let $L(x) = a + bx$ be an arbitrary tangent line to g at μ. 4. Because $L(x)$ is tangent to g at μ, then $g(\mu) = a + b\mu$. Since $\mathbb{P}[X = \mu] = 1$, then $\mathbb{P}[g(X) = g(\mu)] = 1$, $\mathbb{P}[a + bX = a + b\mu] = 1$. Since $g(\mu) = a + b\mu$, then $\mathbb{P}[g(X) = a + bX] = 1$. This holds for every line $a + bx$ tangent to g at μ. Thus by theorem, $\mathbb{E}[g(X)] = g(\mathbb{E}[X])$.

□

Theorem 8 (The Donsker-Varadhan Representation)**.**

For any two probability measures P and Q on a measurable space $(\mathcal{X}, \mathcal{F})$, assume $P \ll Q$ and $D_{\mathrm{KL}}(P\|Q) < \infty$. Then

$$D_{\mathrm{KL}}(P\|Q) = \sup_f \left(\mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f]\right)$$

where the supremum is taken over measurable extended-real-valued functions $f : \mathcal{X} \to [-\infty, \infty)$ such that f is integrable under P and $0 < \mathbb{E}_Q[e^f] < \infty$.

We use the convention $e^{-\infty} = 0$

Proof.

We prove it by proving

- $\mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f] \leq D_{\mathrm{KL}}(P\|Q)$ for all f,
- the equality is achieved for some specific f.

Assume $D_{\mathrm{KL}}(P\|Q) < \infty$, and so $P \ll Q$. Let $p = \frac{dP}{dQ}$ be the Radon-Nikodym derivative.
Consider the quantity $\mathbb{E}_P[f] - D_{\mathrm{KL}}(P\|Q)$. We can write this as a single integral with respect to P:

$$\mathbb{E}_P[f] - D_{\mathrm{KL}}(P\|Q) = \int_{\mathcal{X}} f dP - \int_{\mathcal{X}} \log(p) dP = \int_{\mathcal{X}} \log(e^f) dP - \int_{\mathcal{X}} \log(p) dP = \int_{\mathcal{X}} \log\left(\frac{e^f}{p}\right) dP$$

Since log is concave, by Jensen's inequality, ($\mathbb{E}[\log(Z)] \leq \log(\mathbb{E}[Z])$):

$$\int_{\mathcal{X}} \log\left(\frac{e^f}{p}\right) dP \leq \log\left(\int_{\mathcal{X}} \frac{e^f}{p} dP\right)$$

By Radon-Nikodym, $dP = pdQ$,
Let $S = \{p > 0\}$, then

$$\int_{\mathcal{X}} \frac{e^f}{p} dP = \int_S \frac{e^f}{p} dP = \int_S e^f dQ \leq \int_{\mathcal{X}} e^f dQ$$

and note

$$\log\left(\int_{\mathcal{X}} e^f dQ\right) = \log \mathbb{E}_Q[e^f]$$

Thus

$$\mathbb{E}_P[f] - D_{\mathrm{KL}}(P\|Q) \leq \log \mathbb{E}_Q[e^f]$$

Which is

$$\mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f] \leq D_{\mathrm{KL}}(P\|Q)$$

It is clear that if the equality in the Jensen's inequality is attained, then we are done. Note that the function $\phi(t) = -\log(t)$ is strictly convex and finite in $t > 0$ (log is strictly concave).

By the above corollary, the equality holds iff $Z = \frac{e^{f^*}}{p} = c$ almost surely, where c is the mean of Z and f^* is our optimal function. Then

$$\frac{e^{f^*}}{p} = c \implies e^{f^*} = cp \implies f^* = \log p + \log c = \log\left(\frac{dP}{dQ}\right) + \log c$$

Substitute this optimal function f^* back into the objective:

$$\begin{aligned}
&\mathbb{E}_P[f^*] - \log \mathbb{E}_Q[e^{f^*}] \\
=&\mathbb{E}_P[\log(p) + \log(c)] - \log \mathbb{E}_Q[e^{\log(p)+\log(c)}] \\
=&\mathbb{E}_P[\log(p)] + \log(c) - \log\left(c \cdot \mathbb{E}_Q[p]\right) \\
=&D_{\mathrm{KL}}(P||Q) + \log(c) - \log(c) - \log(1) \\
=&D_{\mathrm{KL}}(P||Q)
\end{aligned}$$

where

$$\mathbb{E}_Q[p] = 1$$

since $pdQ = dP$ and P is a probability measure, so $\int_{\mathcal{X}} pdQ = \int_{\mathcal{X}} dP = 1$.

Therefore the sup is attained by f^*.
Note that from the derivation above, such f^* is not unqiue, we can take any valid c we want. We can take $c = 1$.

□

Remark 9.

The following is a more intuitive proof for DV.
Looking at $\mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f]$, the second term seems not easy to deal with.
If we look at it, which is

$$\log \int e^f \, dQ$$

It has an unnormalized measure

$$e^f \, dQ$$

which has a total mass of

$$Z_f := \int e^f \, dQ = \mathbb{E}_Q[e^f]$$

Normalize it we get a measure

$$dQ_f = \frac{e^f}{Z_f} \, dQ$$

Thus

$$\log \frac{dQ_f}{dQ} = f - \log Z_f$$

And so

$$\mathbb{E}_P[f] - \log Z_f = \mathbb{E}_P[f - \log Z_f] = \mathbb{E}_P \left[\log \frac{dQ_f}{dQ} \right]$$

This is very close to a KL divergence!

Proof.
Let

$$p := \frac{dP}{dQ}.$$

Fix an admissible function f, and define

$$Z_f := \mathbb{E}_Q[e^f]$$

By assumption,

$$0 < Z_f < \infty.$$

Define a new measure Q_f by

$$Q_f(A) := \frac{1}{Z_f} \int_A e^{f(x)} \, dQ(x), \qquad A \in \mathcal{F}.$$

This is a probability measure because

$$Q_f(\mathcal{X}) = \frac{1}{Z_f} \int_{\mathcal{X}} e^{f(x)} \, dQ(x) = 1.$$

By the definition of the Radon–Nikodym derivative,

$$\frac{dQ_f}{dQ} = \frac{e^f}{Z_f}$$

Since

$$\frac{dP}{dQ} = p$$

and

$$\frac{dQ_f}{dQ} = \frac{e^f}{Z_f},$$

the chain rule for Radon–Nikodym derivatives gives

$$\begin{aligned}\frac{dP}{dQ_f} &= \frac{dP/dQ}{dQ_f/dQ} \\ &= \frac{p}{e^f/Z_f} \\ &= pZ_f e^{-f}\end{aligned}$$

Q_f-almost surely hence also P-almost surely because $P \ll Q_f$.

Taking logarithms,

$$\begin{aligned}\log \frac{dP}{dQ_f} &= \log\left(pZ_f e^{-f}\right) \\ &= \log p + \log Z_f - f.\end{aligned}$$

Therefore,

$$\begin{aligned}D_{\mathrm{KL}}(P\|Q_f) &= \mathbb{E}_P\left[\log \frac{dP}{dQ_f}\right] \\ &= \mathbb{E}_P[\log p + \log Z_f - f] \\ &= \mathbb{E}_P[\log p] + \log Z_f - \mathbb{E}_P[f] \\ &= D_{\mathrm{KL}}(P\|Q) + \log Z_f - \mathbb{E}_P[f].\end{aligned}$$

We obtain

$$\mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f] = D_{\mathrm{KL}}(P\|Q) - D_{\mathrm{KL}}(P\|Q_f)$$

By the nonnegativity of KL divergence,

$$D_{\mathrm{KL}}(P\|Q_f) \geq 0$$

Applying this gives

$$\mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f] \leq D_{\mathrm{KL}}(P\|Q)$$

Since this holds for every admissible f,

$$\sup_f \left\{\mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f]\right\} \leq D_{\mathrm{KL}}(P\|Q)$$

The equality holds precisely when

$$D_{\mathrm{KL}}(P\|Q_f) = 0$$

Since

$$D_{\mathrm{KL}}(P\|Q_f) = 0 \quad \Longleftrightarrow \quad Q_f = P$$

if and only if

$$\frac{dQ_f}{dQ} = \frac{dP}{dQ}$$

Therefore,

$$\frac{e^{f^\star}}{Z_{f^\star}} = p$$

Equivalently,

$$e^{f^\star} = Z_{f^\star} p$$

Taking logarithms gives

$$\begin{aligned} f^\star &= \log p + \log Z_{f^\star} \\ &= \log \frac{dP}{dQ} + \log Z_{f^\star} \end{aligned}$$

Since $\log Z_{f^\star}$ is a constant, every maximizer must have the form

$$f^\star = \log \frac{dP}{dQ} + C$$

Now we verify that this function attains the bound.
Fix $C \in \mathbb{R}$, then

$$e^{f^\star} = e^C p$$

Hence

$$\begin{aligned} Z_{f^\star} &= \mathbb{E}_Q[e^{f^\star}] \\ &= \int_{\mathcal{X}} e^C p \, dQ \\ &= e^C \int_{\mathcal{X}} \frac{dP}{dQ} \, dQ \\ &= e^C \int_{\mathcal{X}} dP \\ &= e^C \end{aligned}$$

Therefore

$$\frac{dQ_{f^\star}}{dQ} = \frac{e^{f^\star}}{Z_{f^\star}} = \frac{e^C p}{e^C} = p = \frac{dP}{dQ}.$$

Thus

$$Q_{f^\star} = P$$

Substituting $f^\star$ gives

$$\begin{aligned} \mathbb{E}_P[f^\star] - \log \mathbb{E}_Q[e^{f^\star}] &= \mathbb{E}_P[\log p + C] - \log(e^C) \\ &= \mathbb{E}_P[\log p] + C - C \\ &= D_{\mathrm{KL}}(P \| Q). \end{aligned}$$

Consequently,

$$\sup_f \left\{ \mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f] \right\} \geq D_{\mathrm{KL}}(P \| Q)$$

Combining above we conclude that

$$\sup_f \left\{ \mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f] \right\} = D_{\mathrm{KL}}(P \| Q)$$

This completes the proof.

□

Proof of Theorem 3.2, first part

Theorem 10.

Let $h^\star_{\mathrm{gum},\Delta} : [0,1] \mapsto \mathbb{R}$ be defined as $h^\star_{\mathrm{gum},\Delta}(r) = \log \left(\left\lfloor \frac{1}{1-\Delta} \right\rfloor r^{\frac{\Delta}{1-\Delta}} + r^{\frac{\tilde{\Delta}}{1-\tilde{\Delta}}} \right)$ with $\widetilde{\Delta} = (1-\Delta) \left\lfloor \frac{1}{1-\Delta} \right\rfloor$.
This function gives the optimal $\mathcal{P}_\Delta$-efficiency rates in the sense that $R_{\mathcal{P}_\Delta}(h^\star_{\mathrm{gum},\Delta}) \geq R_{\mathcal{P}_\Delta}(h)$ for any measurable function h.

Moreover, $R_{\mathcal{P}_\Delta}(h^\star_{\text{gum},\Delta})$ is attained at the following least-favorable NTP distribution in $\mathcal{P}_\Delta$:

$$\boldsymbol{P}^\star_\Delta = \left(\underbrace{1-\Delta, \dots, 1-\Delta}_{\lfloor \frac{1}{1-\Delta} \rfloor \text{ times}}, 1-(1-\Delta)\cdot\left\lfloor \frac{1}{1-\Delta} \right\rfloor, 0, \dots \right)$$

The paper proves $R_{P_\Delta}\left(h^\star_{\text{gum},\Delta}\right) \geq R_{P_\Delta}(h)$ by doing the following:

Recall from "optimality via minimax optimization", finding the optimal h that gives $\sup_h R_{\mathcal{P}}(h)$ is reduced to solving $\min_h \max_{P\in\mathcal{P}} L(h,P)$ where $L(h,P) := \mathbb{E}_0\, h(Y) \;+\; \log \mathbb{E}_{1,P}\, e^{-h(Y)}$.

The paper shows

$$\min_h \max_{P\in\mathcal{P}} L(h,P) \geq -D_{KL}(\mu_0, \mu_1, P^\star_\Delta)$$

But the paper also shows the min is attained at

$$\max_{P\in\mathcal{P}_\Delta} L\left(h^\star_{\text{gum},\Delta}, P\right) = -D_{KL}(\mu_0, \mu_1, P^\star_\Delta)$$

And so this $h^\star_{\text{gum},\Delta}$ is the (*unique*) score function that solves the minimax problem

The following are the technical details:

We first claim that the radon-nikodym derivative $\frac{\mathrm{d}\mu_{1,\boldsymbol{P}^\star_\Delta}}{\mathrm{d}\mu_0}$ exists, that is, $\mu_{1,\boldsymbol{P}^\star_\Delta} \ll \mu_0$.

Proof.
It is clear that μ_0, which is Y_t^{gum} under H_0, follows a $U(0,1)$ distribution. Hence μ_0 is the Lebesgue measure on $[0,1]$. Thus for any mble set A, $\mu_0(A) = \int_{A\cap[0,1]} 1\, dr = 0$ implies $A \cap [0,1]$ has length 0.

$\mu_{1,\boldsymbol{P}^\star_\Delta}$, which is the distribution of Y_t^{gum} under H_1 at the "least-favorable NTP distribution". By lemma 3.1, we have CDF

$$\sum_{w\in\mathcal{W}} P_w r^{1/P_w}$$

for $r \in (0,1)$. It is clear that since $U_{t,w} \sim U(0,1)$, for $r \leq 0$ and $r \geq 1$, the CDF is 0 and 1 respectively. Thus we see the support of $\mu_{1,\boldsymbol{P}^\star_\Delta}$ is between $[0,1]$.

Now

$$\mu_{1,\boldsymbol{P}^\star_\Delta}(A) = \int_{A\cap[0,1]} f_1(r)\, dr$$

where $f_1(r)$ is the density of $\mu_{1,\boldsymbol{P}^\star_\Delta}$ distribution (we will derive it later).

Therefore, if $\mu_0(A) = 0$, then $\mu_{1,\boldsymbol{P}^\star_\Delta}(A) = 0$, hence $\mu_{1,\boldsymbol{P}^\star_\Delta} \ll \mu_0$.

$\square$

Thus we can define the radon-nikodym derivative

$$\frac{\mathrm{d}\mu_{1,\boldsymbol{P}^\star_\Delta}}{\mathrm{d}\mu_0}$$

Consider equation 21:
To apply the Donsker-Varadhan representation (we are allowed to do so because the radon-nikodym derivative exists), use the substitution $f = -h$:

$$\begin{aligned}\min_h \left(\mathbb{E}_0[h(Y)] + \log \mathbb{E}_{1,\boldsymbol{P}}[e^{-h(Y)}]\right) &= \min_f \left(\mathbb{E}_0[-f(Y)] + \log \mathbb{E}_{1,\boldsymbol{P}}[e^{f(Y)}]\right) \\ &= -\max_f \left(\mathbb{E}_0[f(Y)] - \log \mathbb{E}_{1,\boldsymbol{P}}[e^{f(Y)}]\right)\end{aligned}$$

By the Donsker-Varadhan representation (with $P = \mu_0$ and $Q = \mu_{1,\boldsymbol{P}}$),

$$-\max_f \left(\mathbb{E}_{\mu_0}[f] - \log \mathbb{E}_{\mu_{1,\boldsymbol{P}}}[e^f]\right) = -D_{\text{KL}}(\mu_0 \| \mu_{1,\boldsymbol{P}})$$

(It can be proven that $\mu_0 \ll \mu_{1,P}$ just like what we did for $\mu_{1,\boldsymbol{P}_\Delta^\star} \ll \mu_0$, except we need $f_1 > 0$ almost everywhere, indeed $f_1(r) > 0$ for $r \in (0,1)$)

We know what is this f, it is the f^* that we derived before! But we do not know what is $\log\left(\frac{dP}{dQ}\right)$.
By definition,

$$\mu_0(A) = \int_A f_0(r)\,\mathrm{d}r$$

$$\mu_{1,\boldsymbol{P}_\Delta^\star}(A) = \int_A f_1(r)\,\mathrm{d}r$$

where f_0, f_1 are the density function of distributions μ_0 and $\mu_{1,\boldsymbol{P}_\Delta^\star}$.

We wish to find g such that

$$\mu_{1,\boldsymbol{P}_\Delta^\star}(A) = \int_A g(r)\,\mathrm{d}\mu_0$$

From definition, it suffices to find

$$\mu_{1,\boldsymbol{P}_\Delta^\star}(A) = \int_A g(r)\,(f_0(r)\,\mathrm{d}r)$$

Compare with above (because $f_0(r) = 1 \neq 0$),

$$g(r) = \frac{f_1(r)}{f_0(r)}$$

Therefore

$$\frac{\mathrm{d}\mu_{1,\boldsymbol{P}_\Delta^\star}}{\mathrm{d}\mu_0} = \frac{f_1(r)}{f_0(r)}$$

Now back to the proof of theorem 3.2, in equation 21,

As we have seen in the proof of The Donsker-Varadhan Representation, the specific $h = -f$ that makes the equality hold (which is also our desired minimizer h) is (by taking $c = 1$, so $\log c = 0$)

$$h^\star_{\mathrm{gum},\Delta}(r) = \log \frac{\mathrm{d}\mu_{1,\boldsymbol{P}_\Delta^\star}}{\mathrm{d}\mu_0}(r) = \log\left(\frac{f_1(r)}{f_0(r)}\right)$$

Now we need to figure out what that is. It is clear that $f_0(r) = 1$

What is $f_1(r)$?

By lemma 3.1, under H_1, the distribution of Y_t^{gum} given P_t obeys

$$\mathbb{P}_{H_1}(Y_t^{\mathrm{gum}} \leq r \mid P_t) = \sum_{w\in\mathcal{W}} P_{t,w} r^{1/P_{t,w}} \quad \text{for } r \in [0,1]$$

For $P_w = 0$, the corresponding term is defined as 0.

By taking its derivative, we obtain the density:

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\sum_{w\in\mathcal{W}} P_{t,w} r^{1/P_{t,w}}\right) = \sum_{w\in\mathcal{W}} P_w\left(\frac{1}{P_w} r^{\frac{1}{P_w}-1}\right) = \sum_{w\in\mathcal{W}} r^{\frac{1-P_w}{P_w}}$$

By subbing in $\boldsymbol{P}_\Delta^\star$, we can get $f_1(r)$.

Recall

$$\boldsymbol{P}_\Delta^\star = \left(\underbrace{1-\Delta, \cdots, 1-\Delta}_{\lfloor \frac{1}{1-\Delta} \rfloor \text{ times}}, 1-(1-\Delta)\cdot \left\lfloor \frac{1}{1-\Delta} \right\rfloor, 0, \cdots \right)$$
$$= \left(\underbrace{1-\Delta, \cdots, 1-\Delta}_{\lfloor \frac{1}{1-\Delta} \rfloor \text{ times}}, 1-\widetilde{\Delta}, 0, \cdots \right)$$

by defining $\widetilde{\Delta} = (1-\Delta)\left\lfloor \frac{1}{1-\Delta} \right\rfloor$.

Therefore

$$f_1(r) = \left\lfloor \frac{1}{1-\Delta} \right\rfloor r^{\frac{\Delta}{1-\Delta}} + r^{\frac{\tilde{\Delta}}{1-\tilde{\Delta}}}$$

and we get the desired $h^\star_{\text{gum},\Delta}$, which is non-decreasing in r.

And the rest is clear by the paper.

□

Proof of lemma 3.2, 3.3 and lemma 3.4

lemma 3.2:

derivative: Use $r^{1/P_i} = e^{\frac{\ln r}{P_i}}$

lemma 3.3:

Remark 11.

$\mathcal{P}_\Delta$ is compact and convex by definition

Proof.
By definition $\mathcal{P}_\Delta := \{\boldsymbol{P} : \max_{w\in\mathcal{W}} P_w \le 1-\Delta\}$, which is

$$\mathcal{P}_\Delta = \left\{ \boldsymbol{P} \in \mathbb{R}^{|\mathcal{W}|} : \sum_{w\in\mathcal{W}} P_w = 1, \quad 0 \le P_w \le 1-\Delta \quad \text{for all } w \in \mathcal{W} \right\}$$

Verification of convexity is clear.

To prove compactness, since we are in $\mathbb{R}^n$, by Heine-Borel, it suffices to prove closed and bounded.

Closeness:

- Let $A := \left\{ \boldsymbol{P} \in \mathbb{R}^{|\mathcal{W}|} : \sum_{w\in\mathcal{W}} P_w = 1 \right\}$. The function $f(\boldsymbol{P}) = \sum P_w$ is continuous. The set A is the preimage of the closed singleton set $\{1\}$ under f, meaning A is closed.
- For each $w \in \mathcal{W}$, $B_w = \left\{ \boldsymbol{P} \in \mathbb{R}^{|\mathcal{W}|} : P_w \ge 0 \right\}$. The projection function $g_w(\boldsymbol{P}) = P_w$ is continuous. The set B_w is the preimage of the closed interval $[0, \infty)$ under g_w, meaning B_w is closed.
- For each $w \in \mathcal{W}$, $C_w = \left\{ \boldsymbol{P} \in \mathbb{R}^{|\mathcal{W}|} : P_w \le 1-\Delta \right\}$. Using the same continuous projection $g_w(\boldsymbol{P}) = P_w$, C_w is the preimage of the closed interval $(-\infty, 1-\Delta]$, meaning C_w is closed.
  Note

$$\mathcal{P}_\Delta = A \cap \left(\bigcap_{w\in\mathcal{W}} B_w \right) \cap \left(\bigcap_{w\in\mathcal{W}} C_w \right)$$

  Thus it is closed.

Boundedness:

$$\|\boldsymbol{P}\|_2 = \sqrt{\sum_{w\in\mathcal{W}} P_w^2} \leq \sqrt{\sum_{w\in\mathcal{W}} 1^2} = \sqrt{|\mathcal{W}|}$$

□

lemma 3.4:

What does it mean by **permutation invariance**:

$$\phi_h(\boldsymbol{P}) = \int_0^1 e^{-h(r)} dF_{1,P}(r)$$

where $F_{1,P}(r) = \sum_{w=1}^{|\mathcal{W}|} P_w r^{1/P_w}$.

Proof of lemma 3.4:

What does it mean by "It suffices to demonstrate that any point in $\mathcal{P}_\Delta$ is in the convex hull of $\Pi(\boldsymbol{P}_\Delta^\star)$, denoted by $\mathrm{Conv}(\Pi(\boldsymbol{P}_\Delta^\star))$":

We wish to prove

$$\mathcal{P}_\Delta = \mathrm{Conv}(\Pi(\boldsymbol{P}_\Delta^\star))$$

because of the following:

Remark 12.

If a convex set is generated by the convex hull of a set of points V, then the extreme points of that convex set must necessarily belong to this set of points V.

What does “It is easy to find that P is the convex interpolation of $|\mathcal{I}|$ points in $\Pi(P_\Delta^\star)$ whose first N coordinates are the same as $1-\Delta$ and the ith coordinate is $1-N(1-\Delta)$” mean:

Note that from the context, $N = m$.
Let

$$R := 1 - m(1-\Delta)$$

denote the remaining probability mass. For each $i \in \mathcal{I}$, select a point $\boldsymbol{E}_i \in \Pi(\boldsymbol{P}_\Delta^\star)$ whose first m coordinates are equal to $1-\Delta$, whose ith coordinate is equal to R, and whose remaining coordinates are all zero. Assign to $\boldsymbol{E}_i$ the weight

$$w_i := \frac{P_i}{R}.$$

Since $\sum_{i\in\mathcal{I}} P_i = R$, these weights satisfy $w_i \geq 0$ and

$$\sum_{i\in\mathcal{I}} w_i = 1.$$

Thus, they define a valid convex combination.

We now verify coordinate-wise that

$$\boldsymbol{P} = \sum_{i\in\mathcal{I}} w_i \boldsymbol{E}_i.$$

For each of the first m coordinates, the value of this convex combination is

$$\sum_{i\in\mathcal{I}} w_i(1-\Delta) = (1-\Delta)\sum_{i\in\mathcal{I}} w_i = 1-\Delta,$$

which agrees with the corresponding coordinate of $\boldsymbol{P}$. For any $j \in \mathcal{I}$, among the selected points, only $\boldsymbol{E}_j$ has a nonzero jth coordinate. Hence, the jth coordinate of the convex combination is

$$w_j R = \frac{P_j}{R} R = P_j,$$

which again agrees with $\boldsymbol{P}$. Finally, every other coordinate is zero in all the selected points and is also zero in $\boldsymbol{P}$. Therefore,

$$\boldsymbol{P} = \sum_{i \in \mathcal{I}} w_i \boldsymbol{E}_i \in \mathrm{Conv}(\Pi(\boldsymbol{P}_\Delta^\star)).$$

If $R = 0$, then $\boldsymbol{P}$ itself already belongs to $\Pi(\boldsymbol{P}_\Delta^\star)$, so the conclusion follows immediately.

In (a), for

$$\boldsymbol{P} = \theta \boldsymbol{P}_1 + (1-\theta)\boldsymbol{P}_2$$

need a valid θ. On the m-th coordinate,

$$P_m = \theta(1-\Delta) + (1-\theta)\big(P_m + P_{m+1} - (1-\Delta)\big)$$

So

$$\theta = \frac{(1-\Delta) - P_{m+1}}{\big((1-\Delta) - P_m\big) + \big((1-\Delta) - P_{m+1}\big)}$$

which is in (0,1) as $1 - \Delta > P_m \geq P_{m+1}$. Sub in θ to $m+1$ coordinates we also get P_{m+1}.